1. Personalized Recommendations Engine
It's an hybrid model made
Content based filtering(user own preferred meals)
Global popularity (collaborative filtering)
User ratings
Cold start for new user
We didn't used neural Network due to the lack of massive amount of historical actual data

Apriori algorithm (cross selling)
Find the target by its id and look for recurrent items alongside it in other billdetails and count how many times those recurrent items appears

2. Natural language processing NLP chatbot

It's a field that helps computers understand, interpret, manipulate, and generate human language.

It uses node nlp, The chatbot takes raw user text, analyzes the language, and classifies it into

a predefined "Intent" (e.g., restaurant.recommend vegan", "restaurant. check_dish_ingredient). It then queries the database based on that intent.
Turn all the intent into a mathematical model using
Tokenization:

Stemming:

Neural Network :

3.Demand forecasting(time series analysis) :

1. **The SQL Aggregation (Data Gathering):**
2. The database does the heavy lifting. It examines all completed orders over a sequential lookback window ($W = 30$ days) and calculates:
    - total_qty: Total units of the item sold over

the window.

- active_days: The number of unique calendar dates the item was ordered.

3. **The Mathematics (Time Series Formulation):**
4. The Time Series algorithm computes the sequential daily sales velocity over the 30-day window, then projects expected demand across a weekly operational replenishment horizon ($H = 7$ days).

const velocity = totalQty / lookbackDays;

// 7-Day Demand Forecast (Y_7d) = V_daily * H
const predictedPeriodQty = velocity * forecastDays;

3. **The Confidence Indicator (Activity Regularity Ratio):**

A time series prediction is only as reliable as the consistency of its underlying observations. If a restaurant sells 30 burgers,

but they were all ordered on a single day for a private party, projecting 1 burger/day across the next week is risky.

To address this, the algorithm calculates an **Activity Regularity Ratio (AR)** :

```javascript
const activityRatio = activeDays / lookbackDays; // Range: [0, 1]
```

- **High Confidence:**activityRatio >= 0.5 (Sold on $\geq 50\%$ of days, highly regular time series).
- **Medium Confidence:**activityRatio >= 0.2 (Sold on $\geq 20\%$ of days, moderate regularity).
- **Low Confidence:**activityRatio < 0.2 (Sold rarely, erratic time series).